

Geospatial Weather and Elevation Analysis

by

Group E

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Introduction and Project Scope

Geospatial Weather and Elevation Analysis is a regional data engineering and exploration project covering 100 fixed locations in Sri Lanka and nearby South and Southeast Asia. It combines hourly weather observations, location metadata, and elevation to support comparisons across places and time. The study profile contains 47 hourly variables from Open-Meteo's ERA5-Seamless reanalysis.

The project brings ingestion, analytical processing, and visualization into one workflow. Users must be able to select weather features, locations, and dates, understand whether the requested data is available, obtain missing observations within source limits, and inspect the resulting charts and analytical summaries. A separate plain-English interface translates weather questions into validated queries and explains the computed results.

1.1 Objectives

- A resumable ingestion process with explicit coverage and quota accounting.
- Preserve source and apply consistent temporal and spatial data rules.
- Organize an analytical warehouse using Snowflake and version-controlled dbt models.
- Serve bounded queries through FastAPI to an interactive React geospatial dashboard.
- Integrate coverage checks, feasible query suggestions, and on-demand downloads.
- Support plain-English analysis with inspectable queries and evidence-based summaries.

1.2 Research Scope

The analysis examines how weather variables vary with elevation and how these relationships vary over time. The coordinates are deliberately selected study sites rather than a representative random sample. Reanalysis values are model estimates, not direct readings from a weather station at every requested coordinate. Comparisons must therefore remain descriptive and account for latitude, geography, sampling, and model resolution.

1.3 Project Design and Implementation Status

The project architecture uses Snowflake and dbt for warehouse storage and transformation. The working local prototype uses Parquet and Python/PySpark for processing and validation, with Arrow-based queries for the interactive map. React/deck.gl, FastAPI, and the plain-English analysis page have local implementations. Warehouse integration and coverage-aware on-demand retrieval are project requirements whose implementation remains outstanding.

2.1 Sources and Geographic Coverage

The location catalogue records a stable identifier, name, country, region, coordinates, and elevation for each of the 100 sites. Coordinates include approximate named locations and mountain sites. Requested coordinates and terrain elevation are retained separately from the grid coordinates and elevation returned by the weather source.

The weather profile requests ERA5-Seamless, a combination of ERA5 and ERA5-Land, with UTC timestamps. It covers thermal and humidity measures, precipitation, wind, pressure, clouds, soil conditions, evapotranspiration, and radiation.

2.2 Temporal Coverage and Ingestion Order

The historical lower bound is 1 January 2020. The active backfill is pinned to 31 August 2026 following the decision to begin with the previous month. It requests the newest seven-day window across all locations before moving to the preceding window. This provides broad geographic coverage early in the download rather than finishing every year for one location first. Windows are clipped at calendar-year boundaries, and verified legacy annual downloads are reused without overlapping observations. The saved study plan defines the exact expected coverage.

Recent reanalysis may be unavailable because the source is delayed. Unavailable data must not be saved as a completed empty checkpoint or confused with an API quota pause. Coverage is assessed from committed observations rather than assumed from the overall earliest and latest dates.

2.3 Quality Rules and Derived Measures

Ingestion checks that each response contains every expected hour exactly once and preserves all requested fields, including typed null values. Each committed chunk has request metadata and a checksum so interrupted runs can resume and existing files can be verified.

Daily values require 24 non-null hourly observations for the selected variable. Daily precipitation is summed, supported continuous measures such as temperature are averaged. Incomplete days remain null.

Findings

3.1 Analytical Scope and Comparable Coverage

The analysis examines data quality, geographic and temporal variation, and the relationship between elevation and weather across the study locations. Source observations are joined with location metadata and aggregated over explicitly verified periods. Coverage checks, consistent aggregation rules, and independent processing validation provide the basis for the findings.

The study locations span 3–4,321 m in elevation. Within Sri Lanka, the 30 sampled sites span 3–2,500 m, while the six coastal sites occupy a narrow 3–10 m band. Comparing these subsets allows the analysis to assess how the available elevation range influences geographic associations.

3.2 Data Quality and Processing Verification

No duplicate location–timestamp keys were found, and the metadata join preserved all 1,230,960 rows without unmatched locations or row multiplication. The 47 numeric weather fields were checked separately from partition metadata.

Fifteen broad plausibility screens found no out-of-range observations. Temperature and surface-pressure bounds accommodate both coastal and mountain locations. No observations were removed by these checks. Temperatures range from -8.0°C to 40.7°C and surface pressure falls to 599.8 hPa. There are 1,606 hourly rows with positive snowfall and 4,567 with positive snow depth.

3.3 Geographic and Temporal Variation

Table 3.1: Pooled weather measures. These describe the selected sites and are not national or annual climate averages.

Measure	Value	Interpretation
Hourly mean temperature	22.80°C	Pooled across selected sites
Relative humidity	83.11%	Pooled hourly mean
Wind speed	2.47 m/s	Pooled hourly mean
Hourly precipitation	0.356 mm	Pooled hourly mean
Mean daily precipitation	8.53 mm	Equivalent under complete 24-hour coverage

Temperature varies substantially across the sampled elevation range

Table 3.2: Highest and lowest sampled site mean temperatures

Rank	Location	Mean temperature
Warmest	Chennai	31.04°C
2	Madurai	30.68°C
3	Batticaloa	30.44°C
Coolest	Nathu La	6.27°C
2	Dingboche	7.66°C
Range	Warmest to coolest	24.77°C

Rainfall also varies substantially between locations. The largest observed period rainfall totals and highest mean wind speeds are shown below

Table 3.3: Leading rainfall totals and mean wind speeds.

Measure	Rank	Location	Value
Rainfall total	1	Lachen	3,387.2 mm
Rainfall total	2	Sylhet	3,147.4 mm
Rainfall total	3	Pokhara	3,108.1 mm
Mean wind speed	1	Jaffna	7.92 m/s
Mean wind speed	2	Mannar	7.79 m/s
Mean wind speed	3	Trincomalee	6.21 m/s

3.4 Elevation Relationships

Table 3.4: Elevation versus location mean temperature.

Sample	Sites	Elevation (m)	Pearson r	Nominal p
All regional sites	100	3–4,321	−0.981	3.81×10^{-71}
Sri Lanka	30	3–2,500	−0.965	8.20×10^{-18}
Coastal subset	6	3–10	+0.685	0.133

This contrast shows the sensitivity of the correlation to geographic selection and elevation range. The coastal subset provides too little elevation variation to characterise the relationship observed across the full study area.

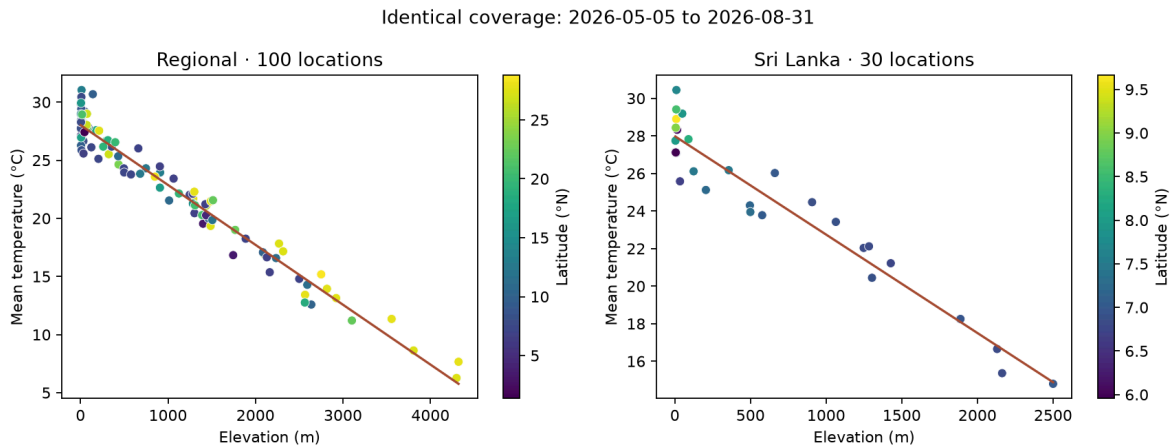


Figure 3.1: Elevation–temperature associations for all regional sites and the Sri Lankan subset.

Table 3.5: Pearson correlations between location-level weather summaries and latitude or elevation.

Measure and predictor	Regional r	Sri Lankan r
Temperature versus elevation	−0.981	−0.965
Mean daily precipitation versus elevation	+0.337	+0.285
Wind speed versus elevation	−0.506	−0.531
Relative humidity versus elevation	+0.418	+0.232
Surface pressure versus elevation	−0.998	−0.9995
Cloud cover versus elevation	+0.312	+0.515
Shortwave radiation versus elevation	−0.229	−0.129

The table shows that elevation has the strongest temperature and surface pressure association in both samples. Elevation’s associations with rainfall and wind are less consistent and do not justify a universal claim that either measure is controlled by elevation, exposure, geography, season, and sample selection remain relevant.

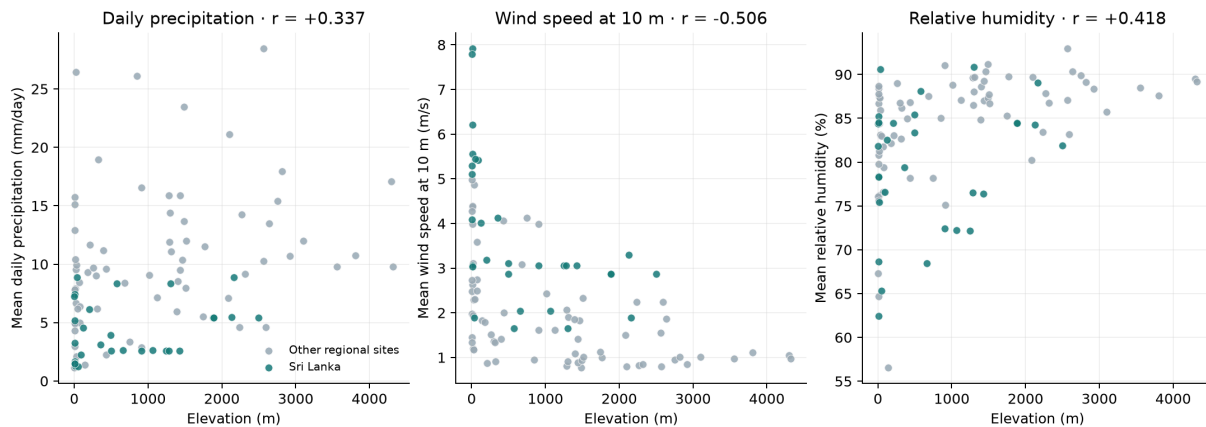


Figure 3.2: Elevation versus mean daily precipitation, wind speed, and relative humidity.

Rainfall differs substantially among sites at similar elevations, indicating that elevation alone provides an incomplete description of precipitation. The humidity relationship also depends on the geographic sample, while wind speed decreases with elevation in both groups despite several low-elevation sites having particularly high wind speeds.

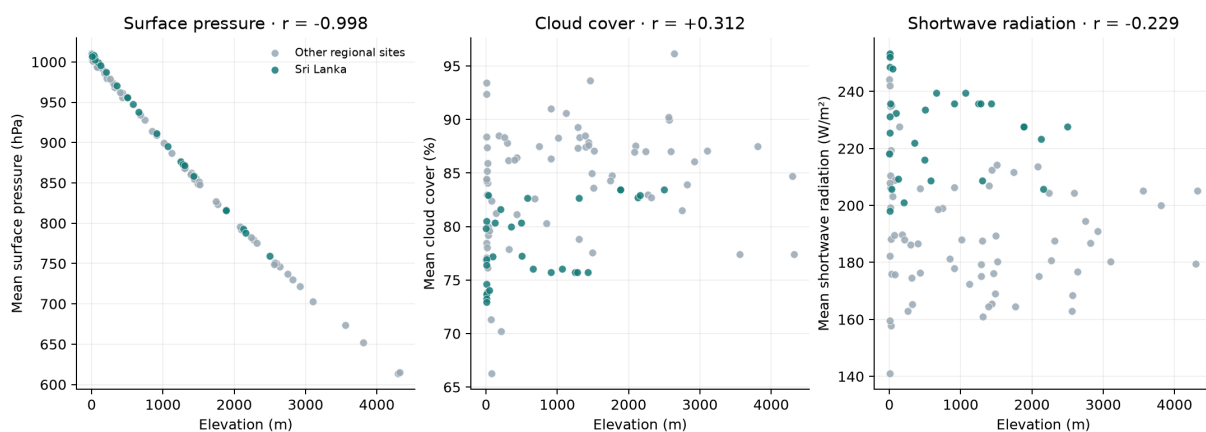


Figure 3.3: Elevation versus surface pressure, cloud cover, and shortwave radiation. Surface pressure is measured at the site and is distinct from pressure reduced to mean sea level.

Surface pressure has a much tighter relationship with elevation than the other measures. Cloud cover is more positively associated with elevation within Sri Lanka than across the region, while shortwave radiation remains weakly related in both samples.

3.4.1 Relationships Between Year and Locations on Weather Variables

The same-month comparisons cover six locations, seven variables, and twelve calendar months: 504 comparisons, each testing both ranks and within-month spread across 2020–2025. There are 43 detected rank differences and 8 detected spread differences. These counts refer to location–variable–month combinations, not individual pairs of years. All six locations have detected differences.

Table 3.6: Number of months with detected rank / spread differences. Ranks are tested with Kruskal–Wallis and spread with Fligner–Killeen scores.

Feature	Batticaloa	Colombo	Galle	Jaffna	Matara	Trincomalee
Temperature	3 / 0	6 / 0	5 / 0	5 / 0	4 / 0	3 / 0
Daily precipitation	0 / 0	0 / 1	0 / 1	0 / 1	1 / 1	0 / 0
Wind speed at 10 m	1 / 0	1 / 0	1 / 0	0 / 1	2 / 0	1 / 0
Relative humidity	0 / 1	3 / 1	3 / 0	1 / 0	3 / 0	0 / 0
Cloud cover	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 1

Every detected combination is visualised below. Boxplots show medians and quartiles, triangles show means, whiskers use 1.5 IQR, and dots show individual daily values.

3.4.2 Batticaloa

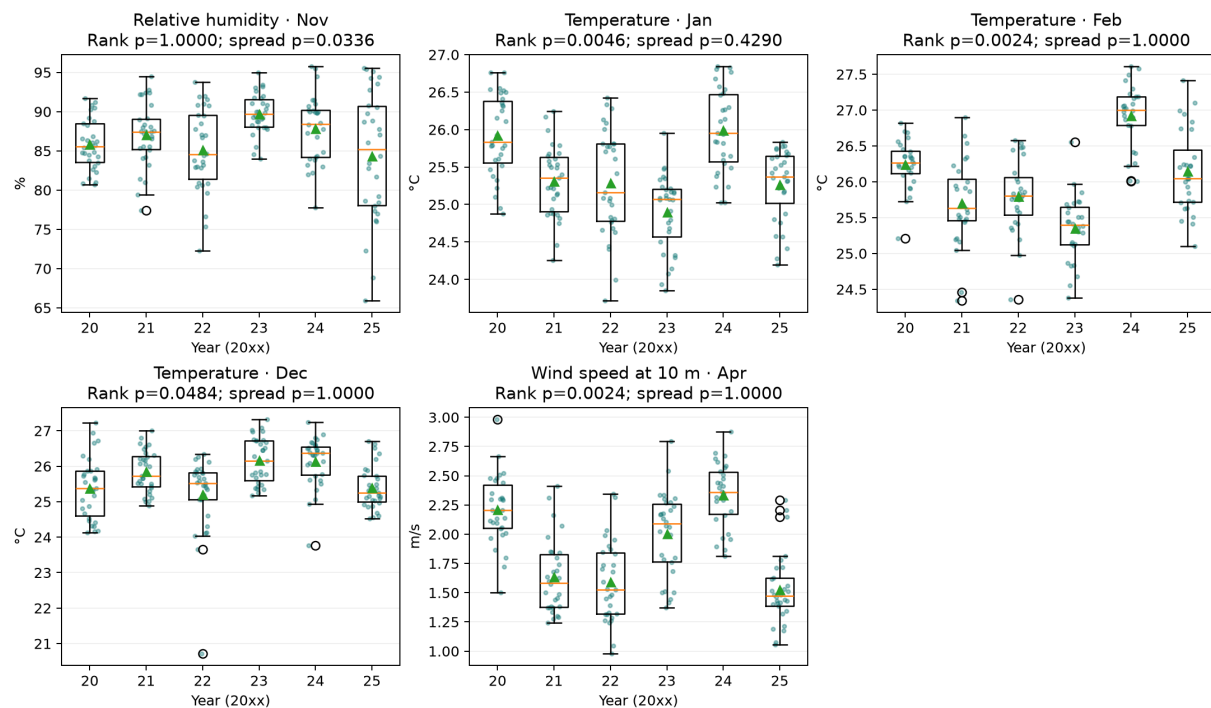


Figure 3.4: Batticaloa: detected differences across 2020–2025.

3.4.3 Colombo

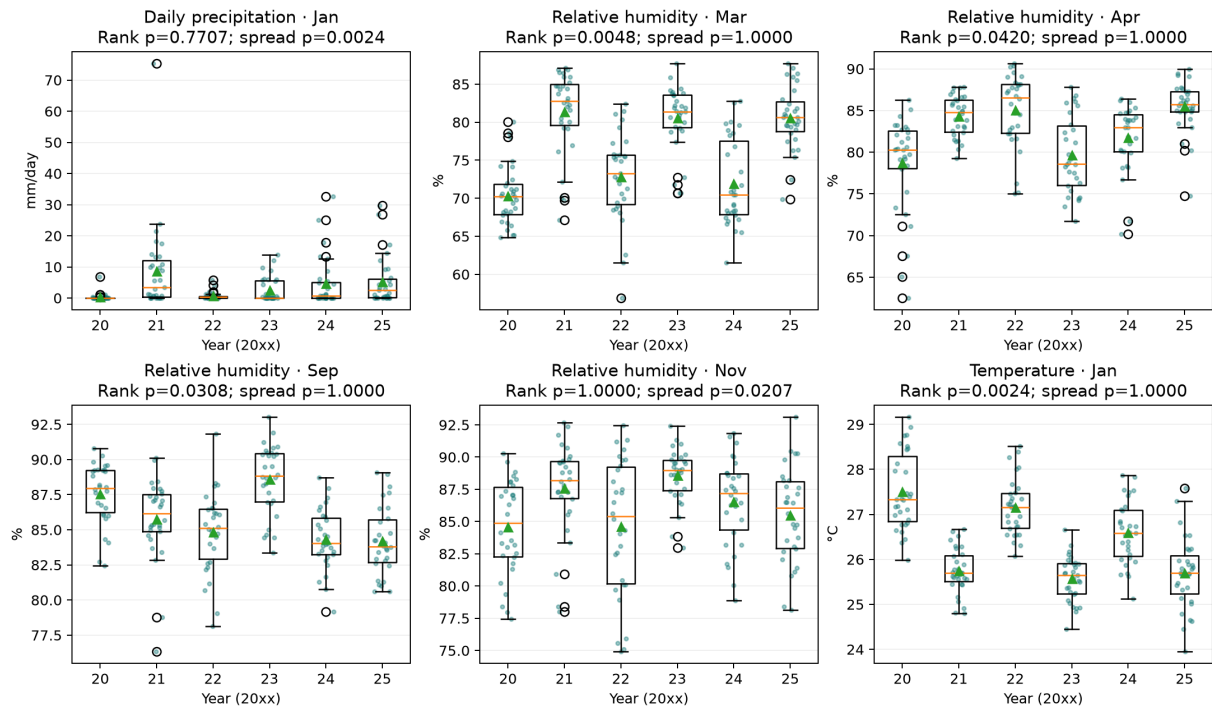


Figure 3.5: Colombo: detected differences across 2020–2025.

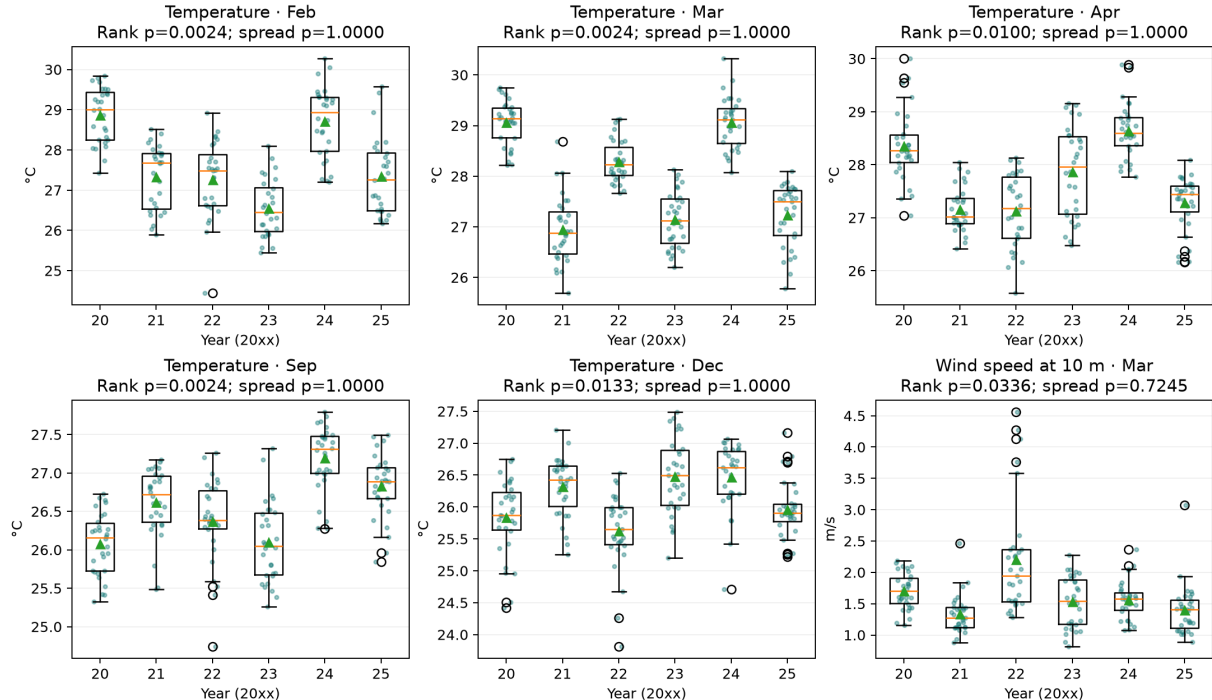


Figure 3.6: Colombo: detected differences across 2020–2025.

3.4.4 Galle

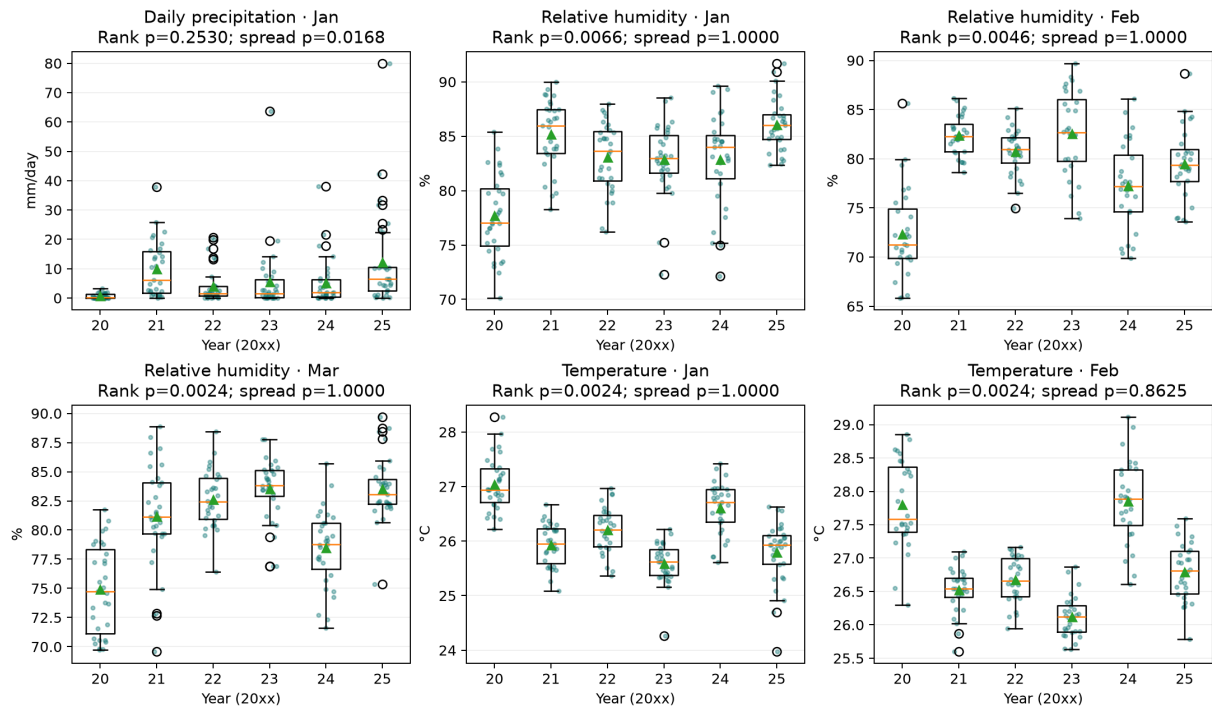


Figure 3.7: Galle: detected differences across 2020–2025.

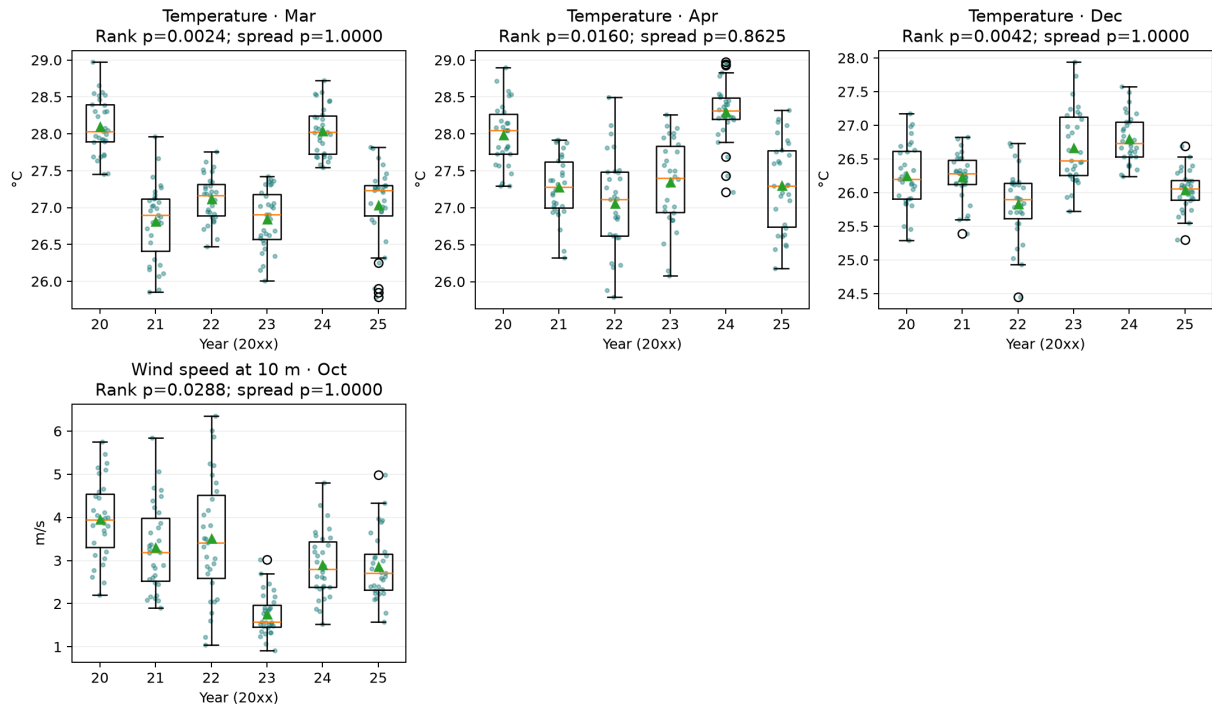


Figure 3.8: Galle: detected differences across 2020–2025.

3.4.5 Jaffna

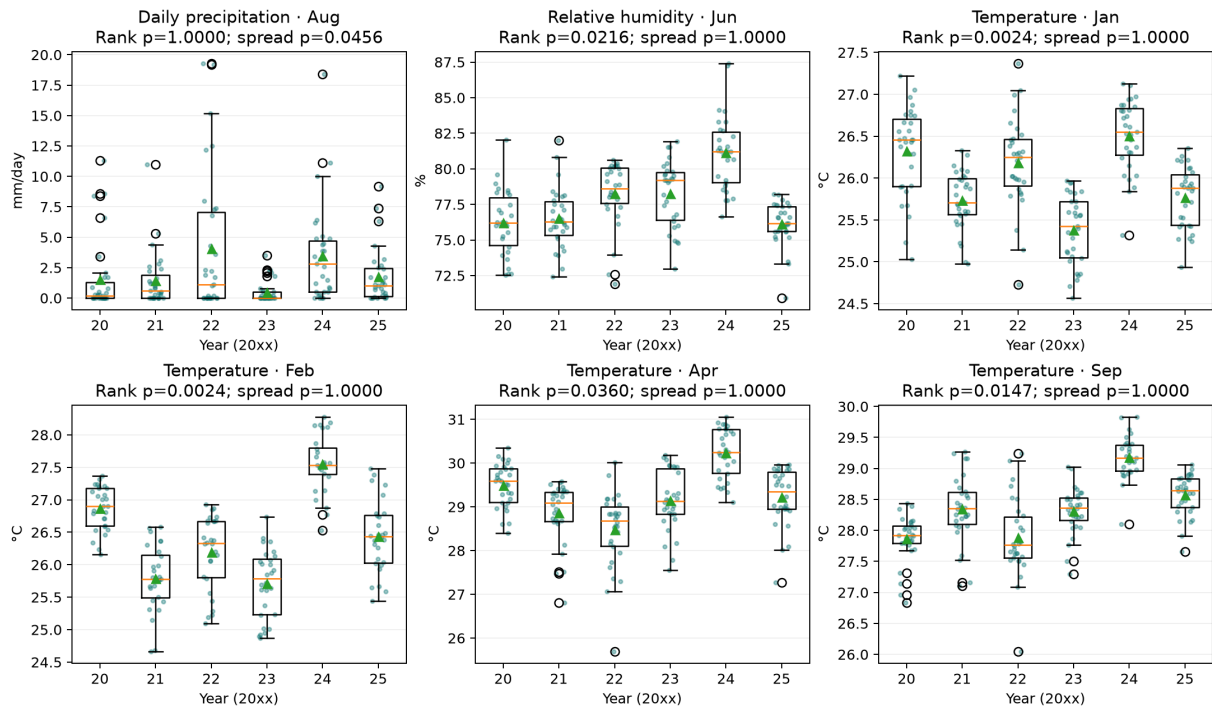


Figure 3.9: Jaffna: detected differences across 2020–2025.

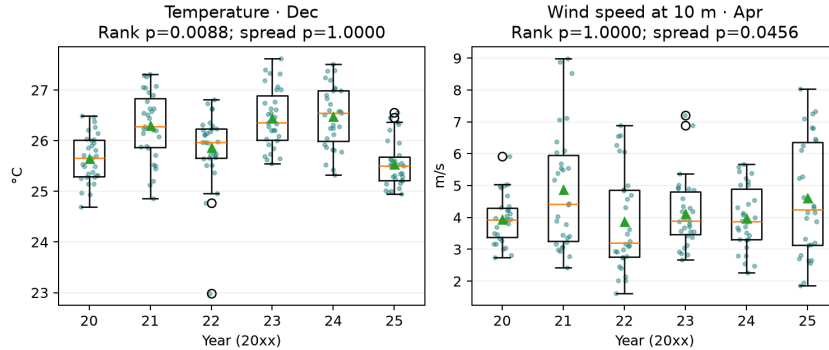


Figure 3.10: Jaffna: detected differences across 2020–2025.

3.4.6 Matara

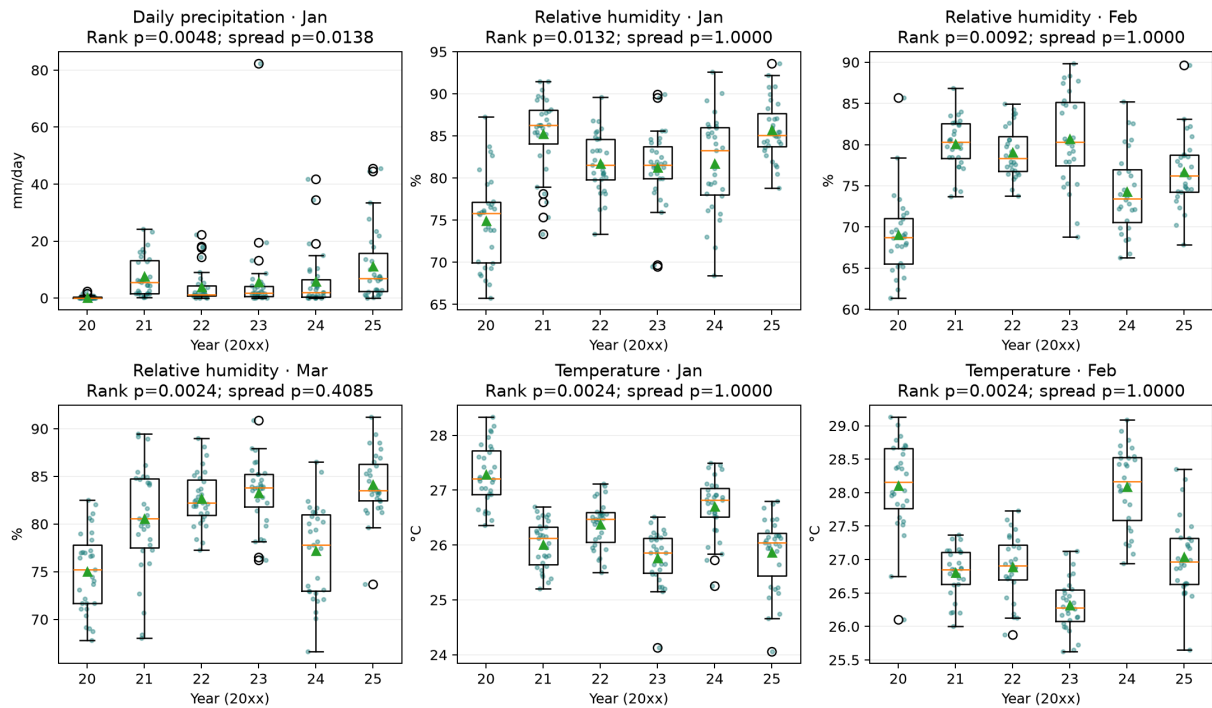


Figure 3.11: Matara: detected differences across 2020–2025.

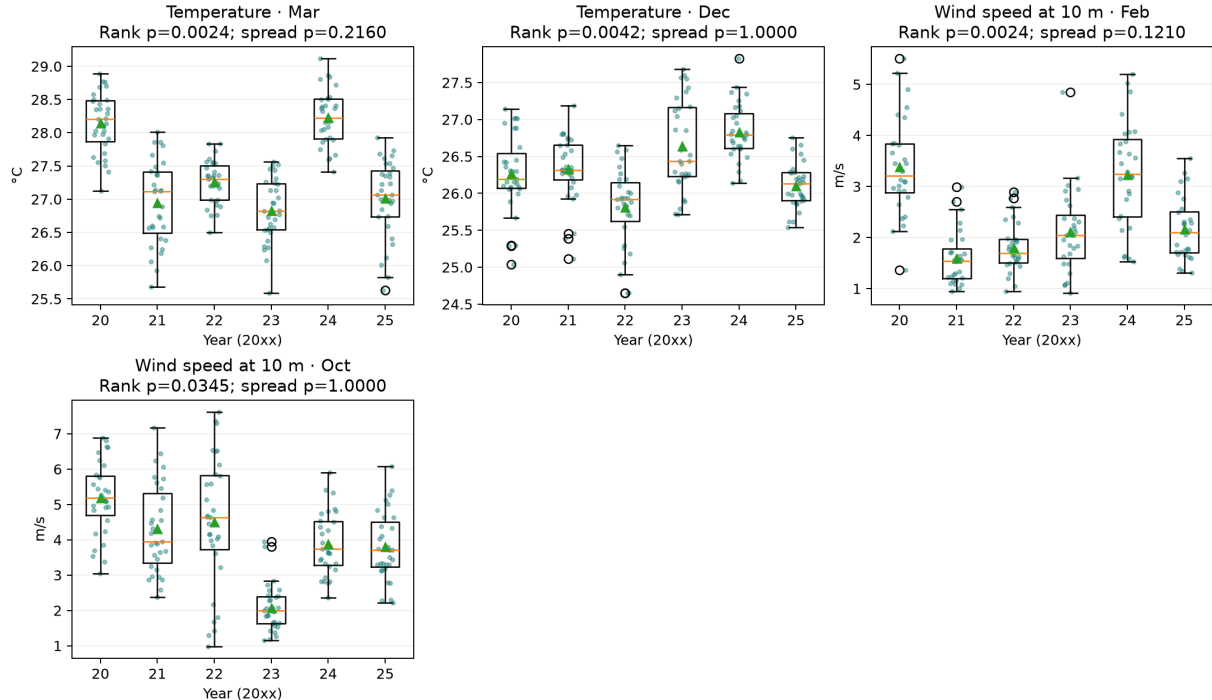


Figure 3.12: Matara: detected differences across 2020–2025.

3.4.7 Trincomalee

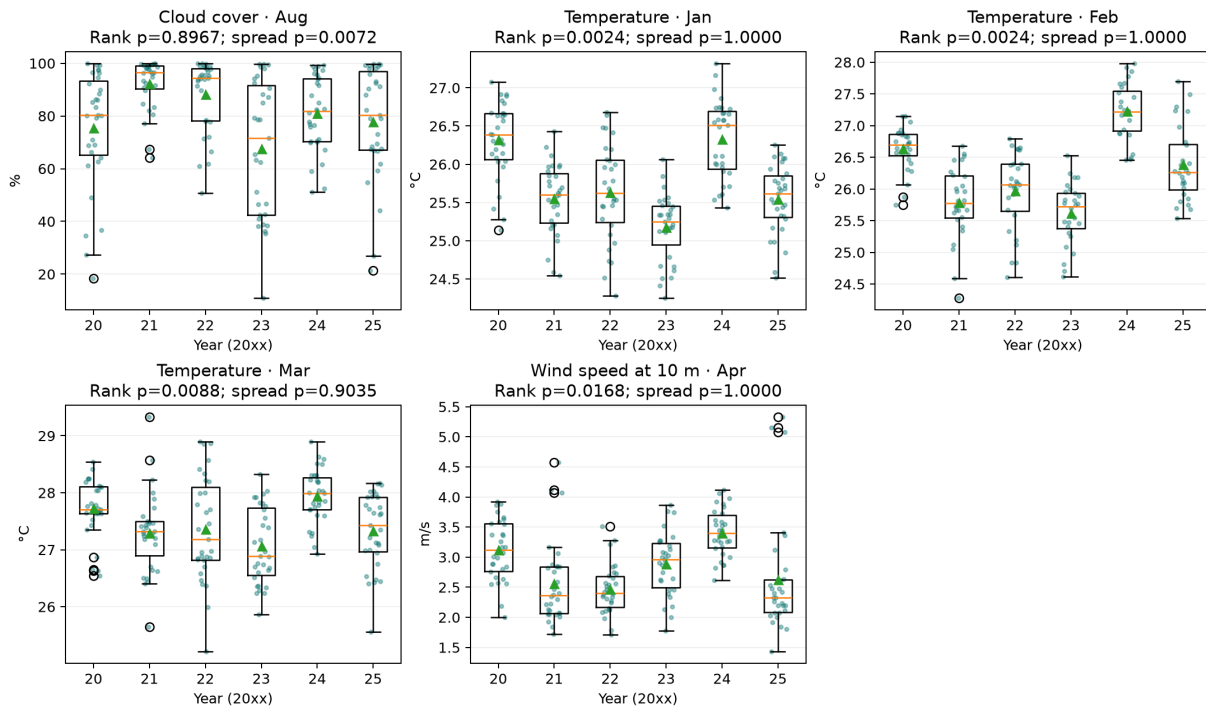


Figure 3.13: Trincomalee: detected differences across 2020–2025.

3.5 Elevation Differences Within Months in 2026

The same 100 locations are compared within each of January–August 2026 using newly committed historical downloads. All eight months are complete, including May. This updated coverage contains 583,200 hourly observations and 800 complete location-month summaries. Source hashes and per-month hourly counts are verified in a separate snapshot, extending the earlier May–August comparison without changing the frozen historical year-by-year analysis.

Locations are divided into three fixed elevation bands: below 500 m (48 sites), 500 to below 1,500 m (26 sites), and at least 1,500 m (26 sites). These thresholds are applied consistently to every feature and month. Kruskal–Wallis compares rank distributions across the three bands separately in each month. The plots show variation between sites within each elevation band, not within-month daily variation. That differs from the historical year-by-year comparisons.

Table 3.7: Elevation-band differences. Values below 0.05 indicate detected differences under country-restricted permutations.

Feature	Jan	Feb	Mar	Apr
Temperature	0.0056	0.0056	0.0056	0.0056
Daily precipitation	0.2760	0.0092	0.0056	0.0056
Wind speed at 10 m	0.0056	0.0056	0.0056	0.0056
Relative humidity	1.0000	1.0000	1.0000	0.4238
Surface pressure	0.0056	0.0056	0.0056	0.0056
Cloud cover	1.0000	0.7344	0.0056	0.0075
Shortwave radiation	1.0000	0.5360	0.5093	1.0000

Table 3.8: Elevation-band differences. Values below 0.05 indicate detected differences under country-restricted permutations.

Feature	May	Jun	Jul	Aug
Temperature	0.0056	0.0056	0.0056	0.0056
Daily precipitation	0.0210	0.0210	0.1003	0.0576
Wind speed at 10 m	0.0056	0.0056	0.0056	0.0056
Relative humidity	0.0266	0.0056	0.0056	0.0056
Surface pressure	0.0056	0.0056	0.0056	0.0056
Cloud cover	1.0000	0.1264	0.0154	0.0056
Shortwave radiation	1.0000	0.4238	0.3640	0.0075

3.5.1 Temperature Across Elevation Bands

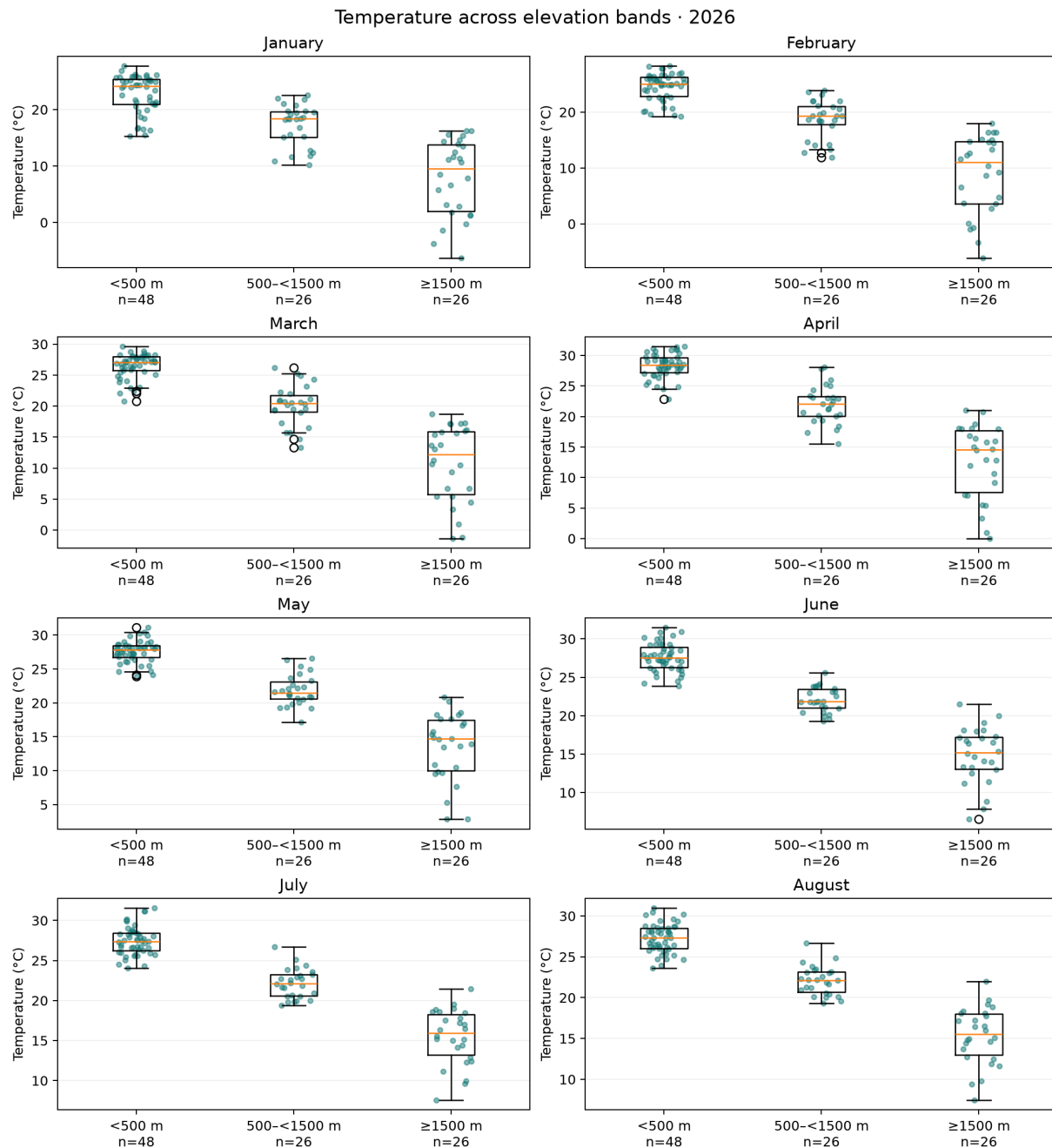


Figure 3.14: Temperature by elevation band in January–August 2026. Dots represent individual location summaries; boxes show medians and interquartile ranges, with 1.5-IQR whiskers.

3.5.2 Daily precipitation Across Elevation Bands

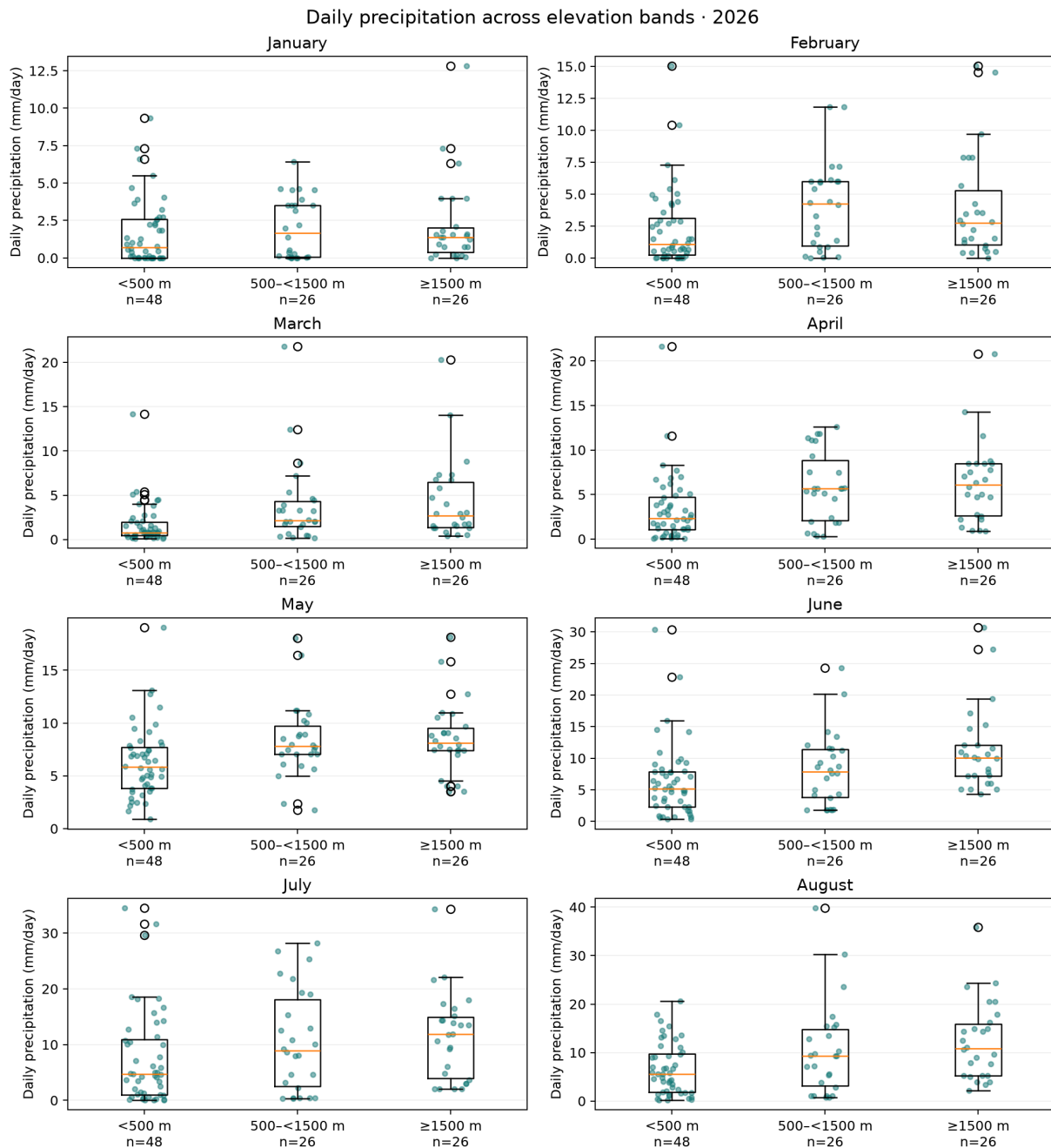


Figure 3.15: Daily precipitation by elevation band in January–August 2026. Dots represent individual location summaries; boxes show medians and interquartile ranges, with 1.5-IQR whiskers.

3.5.3 Wind speed at 10 m Across Elevation Bands

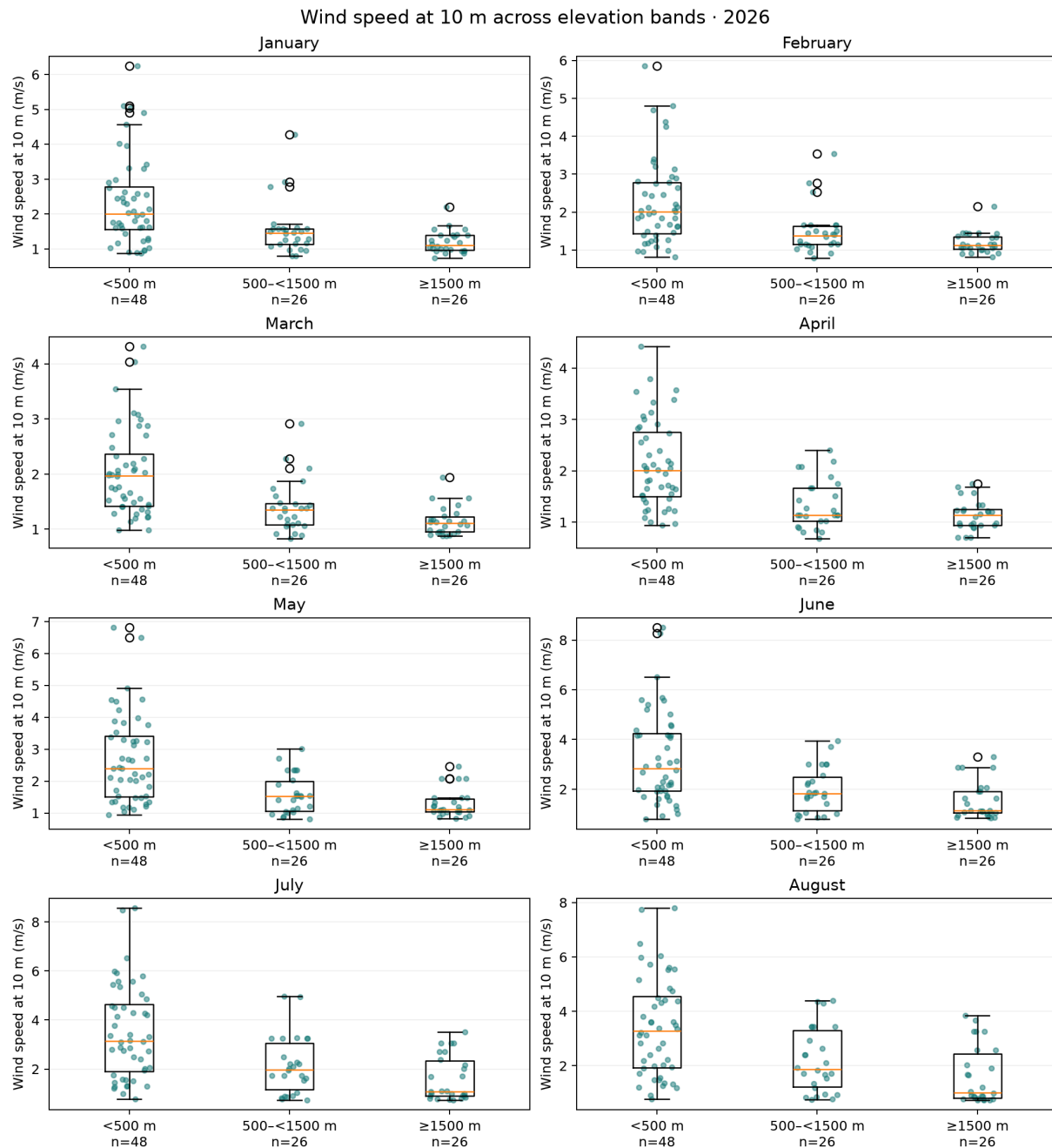


Figure 3.16: Wind speed at 10 m by elevation band in January–August 2026. Dots represent individual location summaries; boxes show medians and interquartile ranges, with 1.5-IQR whiskers.

3.5.4 Relative humidity Across Elevation Bands

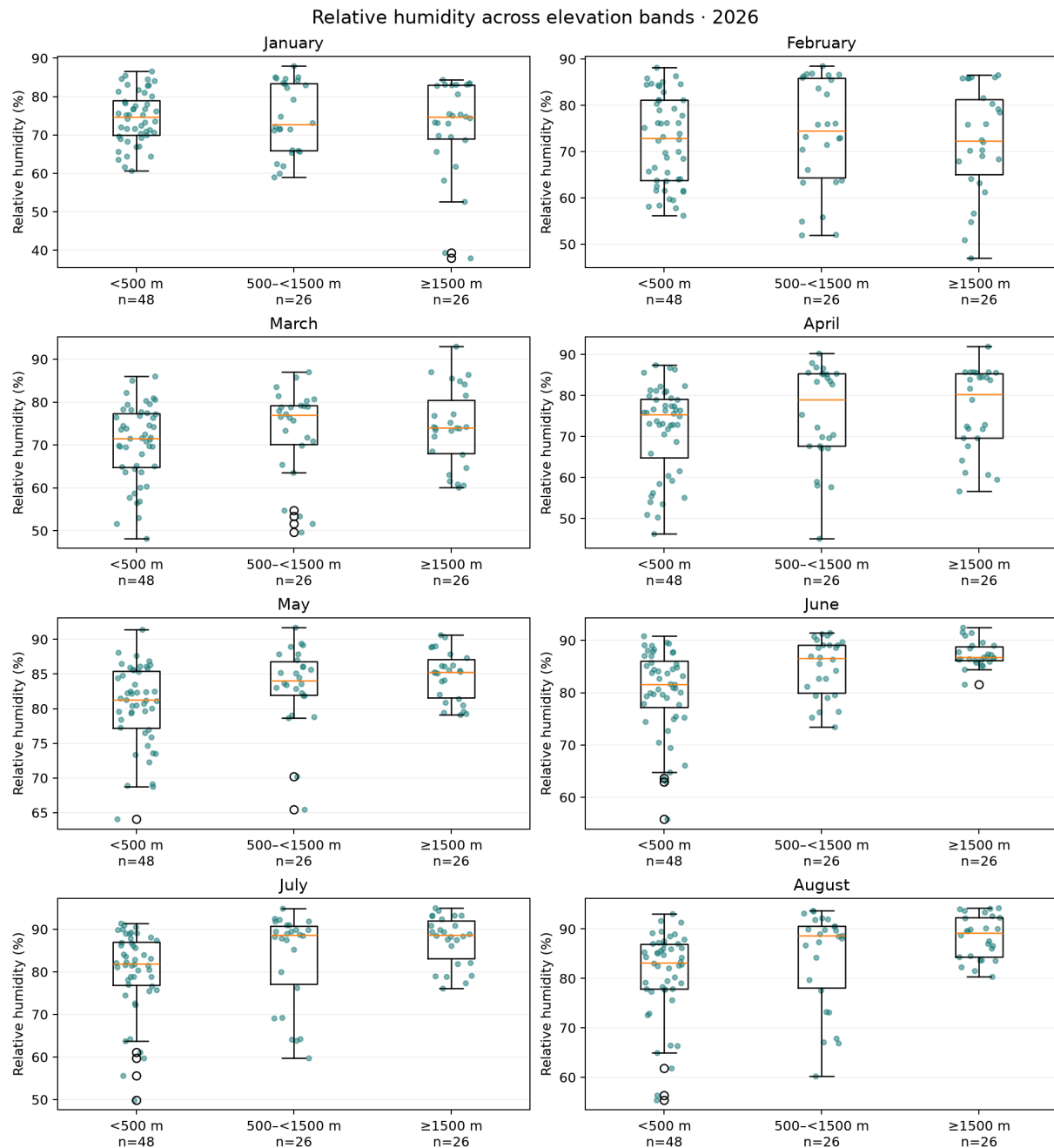


Figure 3.17: Relative humidity by elevation band in January–August 2026. Dots represent individual location summaries; boxes show medians and interquartile ranges, with 1.5-IQR whiskers.

3.5.5 Surface pressure Across Elevation Bands

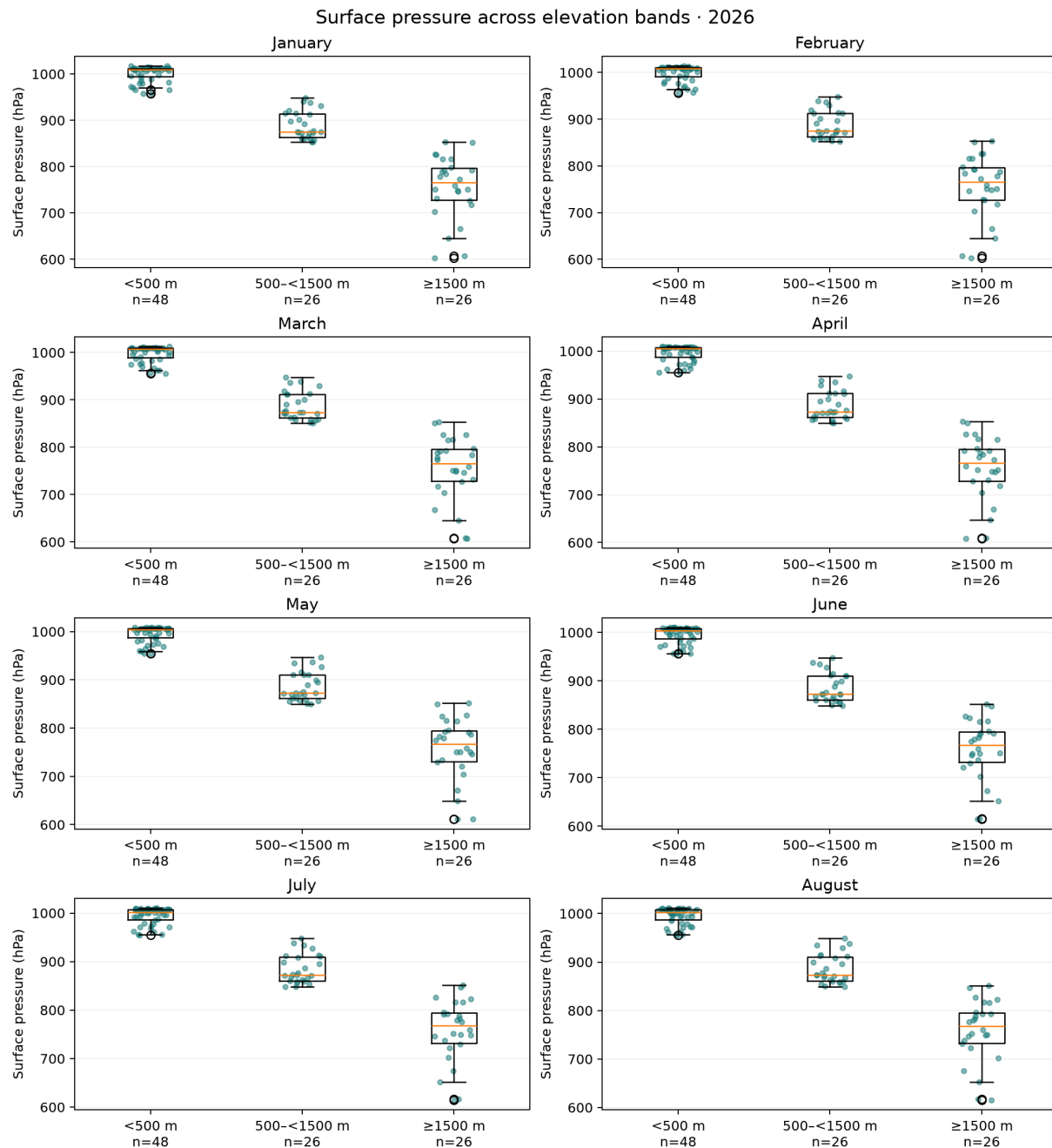


Figure 3.18: Surface pressure by elevation band in January–August 2026. Dots represent individual location summaries; boxes show medians and interquartile ranges, with 1.5-IQR whiskers.

3.5.6 Cloud cover Across Elevation Bands

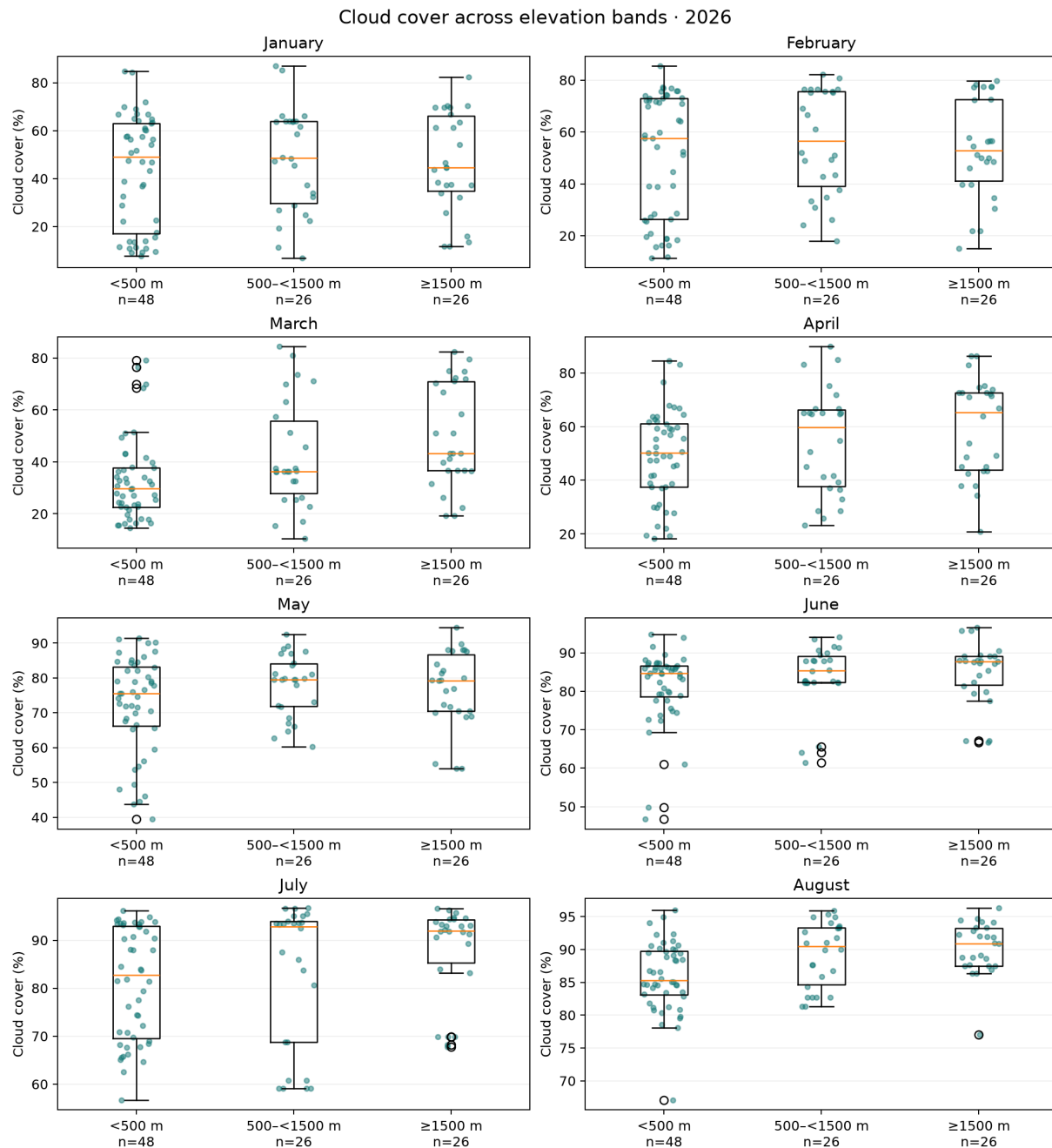


Figure 3.19: Cloud cover by elevation band in January–August 2026. Dots represent individual location summaries; boxes show medians and interquartile ranges, with 1.5-IQR whiskers.

3.5.7 Shortwave radiation Across Elevation Bands

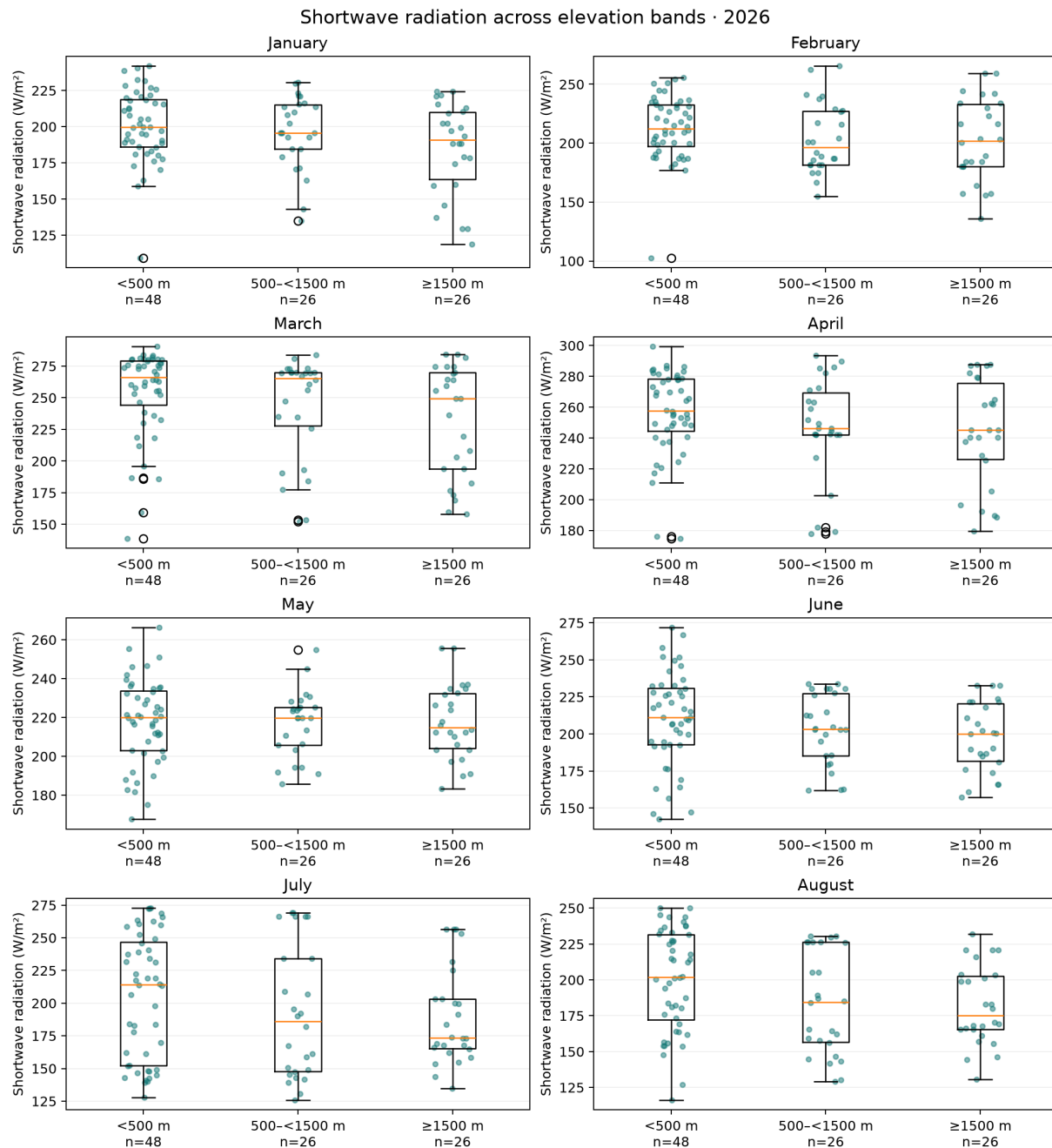


Figure 3.20: Shortwave radiation by elevation band in January–August 2026. Dots represent individual location summaries; boxes show medians and interquartile ranges, with 1.5-IQR whiskers.

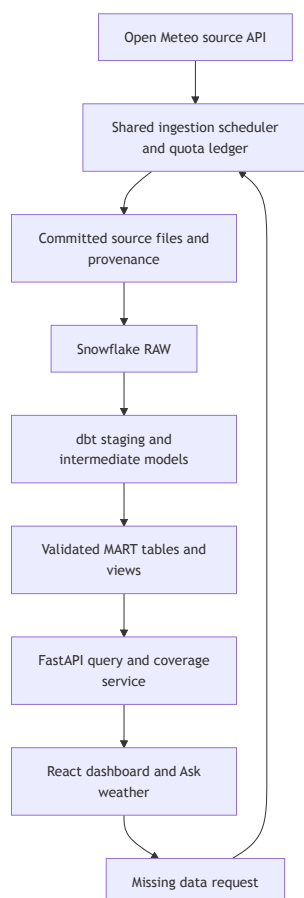


Figure 4.1: On-demand Ingestion

The source retrieval service and the analytical query service have different roles. Ingestion acquires missing observations under the source quota. The serving layer reads validated data and returns only the requested analytical result. An interactive query against stored data does not itself imply that missing source data has been retrieved.

4.1 Technology Selection

4.1.1 Snowflake

Snowflake is selected for the shared warehouse design because it provides managed analytical storage and independently scalable virtual warehouses. Snowflake manages the underlying compute infrastructure, so database managers and data engineers do not need to provision, patch, or manually maintain virtual machines and cluster runtimes. Compute can be suspended when it is not needed and scaled for heavier workloads, allowing transformation jobs and interactive serving to use separate warehouses while sharing one governed analytical dataset. Its SQL interface also fits the location and time filters, aggregations, and window calculations used by the dashboard.

This managed, SQL-first operating model is a strong fit for the project because it supports centralized access, role-based governance, and a clear deployment boundary between ingestion, transformation, and serving. Compared with a Databricks-first design, Snowflake requires less platform setup for this warehouse workload: teams do not need to assemble and maintain clusters, runtime environments, and notebook-based execution as part of the core serving platform. Databricks is powerful and well suited to developer-led data engineering, distributed processing, and data science, but its flexibility can introduce more operational choices and setup than this project needs. A local Parquet/Spark workflow remains sufficient for development and may also be sufficient for the present dataset size. Snowflake is therefore selected for team usability, managed operations, multi-user access, and independent compute scaling, rather than simply because the project contains millions of rows. The trade-offs are cloud dependence, operating cost, role administration, and the need to monitor and control warehouse activity.

4.1.2 Data Marts

A data mart is the curated analytical layer of the warehouse, not another database product. It presents data at a defined grain for a specific use. The project design uses location-day weather records for map playback and history, seasonal or annual summaries for comparisons, and coverage records for availability checks. This follows the separation of reusable staging logic from consumer-facing models described in dbt's guidance on marts.

This layer prevents each dashboard or generated query from redefining the same 24-hour completeness rule, precipitation calculation, units, and location joins. Materializing frequently requested aggregates can reduce repeated computation, views can expose less frequently used results without creating a stored copy of every possible combination. That choice requires measurement rather than assuming all marts are automatically faster.

The raw hourly layer remains available for new analyses and audit. Marts must retain coverage counts and meaningful keys because a summary can hide missing data or unequal sample sizes. Additional models also create refresh dependencies and storage costs. The design therefore starts with the grains actually needed by the dashboard and expands only when a clear analytical requirement warrants it.

4.1.3 dbt

dbt provides a framework for defining transformations as modular models, expressing their dependencies, and documentation alongside the code.

For this project, dbt makes the path from raw hourly observations to validated daily and comparative measures explicit and reviewable. A change to the complete-day rule can be maintained in a shared model rather than copied into multiple API handlers.

Snowflake supplies the storage and execution compute, dbt organizes and runs the transformation logic that produces the marts. It does not replace the Open-Meteo ingestion client, quota scheduler, or FastAPI service. Tests can catch key and coverage violations, but their presence alone does not establish scientific validity. The trade-off is an additional project configuration and deployment layer compared with a small collection of SQL scripts. This is justified when several contributors maintain dependent models and reproducible releases, the team must still design incremental boundaries carefully for backwards-in-time backfills.

4.2 Resumable Ingestion

The ingestion service records the study configuration, completed chunks, source request windows, timestamps, checksums, and quota usage. Atomic writes separate an in-flight response from a committed checkpoint. One writer per study prevents two processes from writing the same chunk. A shared ledger coordinates weighted request costs across runs and must accompany the dataset when ingestion moves to another computer.

4.3 Snowflake and dbt Warehouse Design

The warehouse is organized into RAW, STAGING, INTERMEDIATE, and MART schemas. The raw layer retains source observations and ingestion provenance. Staging models standardize names, types, and timestamps. Intermediate models join the location and elevation catalogue and derive temporal and geographic dimensions. Mart models apply the complete-day rule and expose daily metrics and analytical evidence views.

Source files enter through a controlled landing area and Snowflake stage. The implementation must select and configure the loading mechanism, incremental keys, and recovery boundaries before enabling unattended ingestion. Incremental processing must handle both older backfill windows and newly available observations without creating duplicate location-hours.

dbt tests must cover unique keys, non-null identifiers, accepted values, catalogue relationships, timestamp bounds, and daily observation counts. Release metadata identifies the source coverage, transformation version, and active analytical snapshot. A reconciliation run must compare warehouse outputs with the validated local reference before the dashboard changes its serving source.

Dashboard and User Workflows

5

5.1 Geospatial Exploration

The primary interface uses React, TypeScript, and deck.gl. A sidebar contains the weather feature, country, date range, and Load weather controls. Map settings are opened from a button at the top-right of the chart. The interface uses a plain white background, with the map, coverage information, and selected location taking visual priority.

Elevation controls geographic height and the selected weather feature controls colour. Interpolated colours are applied to the terrain tiles themselves, following hills and valleys. The interpolation uses nearby valid samples, missing vertices, long connections, and regions outside supported triangles remain unfilled. These colours are a visual estimate between samples, not additional measured observations. Display exaggeration changes the view without changing recorded elevation or weather.

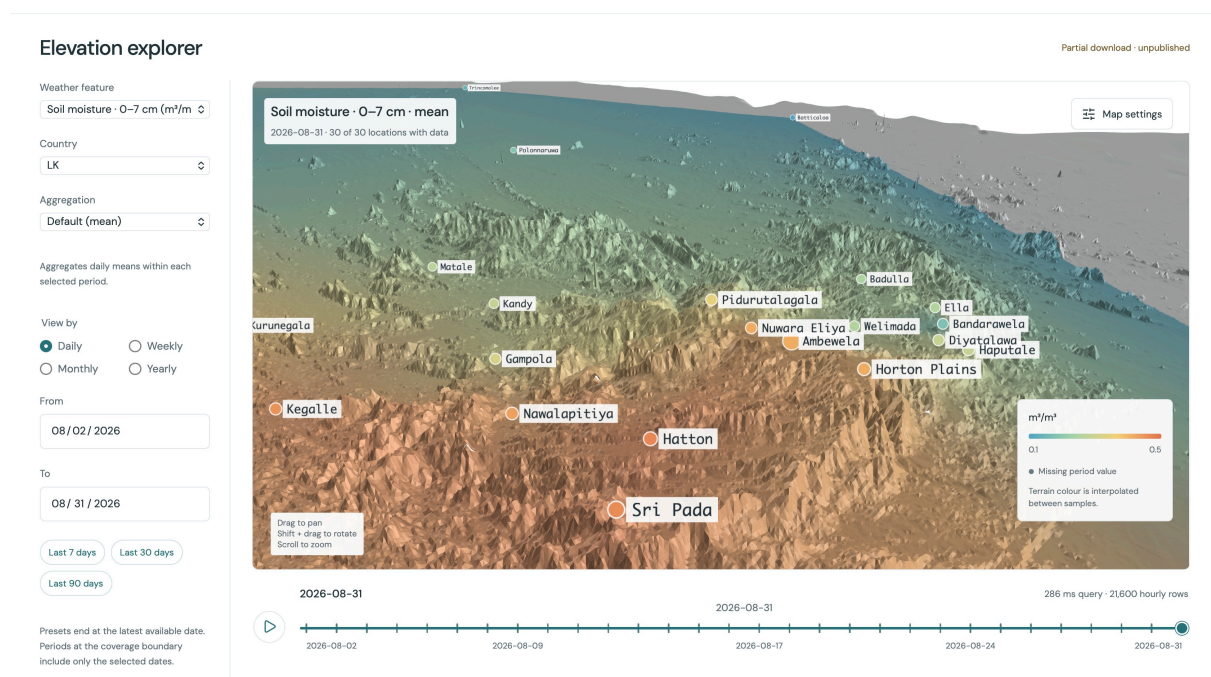


Figure 5.1: The dashboard interface

Users can pan, rotate, zoom, select a location, and inspect its history. The date slider and playback reuse the loaded window. The colour scale remains fixed within that window so changes over time are comparable.

5.2 Query and Response Lifecycle

The current map query is bounded to 100 locations and 93 days. Selecting Load weather requests the chosen feature and window, editing controls alone does not relabel the loaded scene. A newer request cancels the older browser request, and stale responses must not overwrite the current selection. Missing observations remain visibly missing rather than becoming zero.

The local map service filters and aggregates Parquet through Arrow. The warehouse serving design queries validated Snowflake marts through the same application boundary. Responses are currently returned as complete JSON windows. Playback is interactive, but this does not constitute live source ingestion or an implemented Kafka, WebSocket, or server-sent-event pipeline.

5.3 Coverage-aware, On-demand Ingestion

Coverage-aware retrieval is a core dashboard requirement. Before fulfilling a selection, the service must check the requested locations, dates, and weather feature against committed coverage. It must identify missing observations and estimate the required source requests, weighted quota cost, and likely waiting time. Source unavailability and quota exhaustion must be explained separately.

If the full selection cannot be served immediately, the dashboard must calculate practical alternatives: fewer locations with complete coverage, a shorter complete date window, or the available subset with gaps marked. For example, an interface could suggest using 12 complete locations or the latest seven complete days. Such numbers are illustrative, actual suggestions must come from the current coverage catalogue and remaining quota.

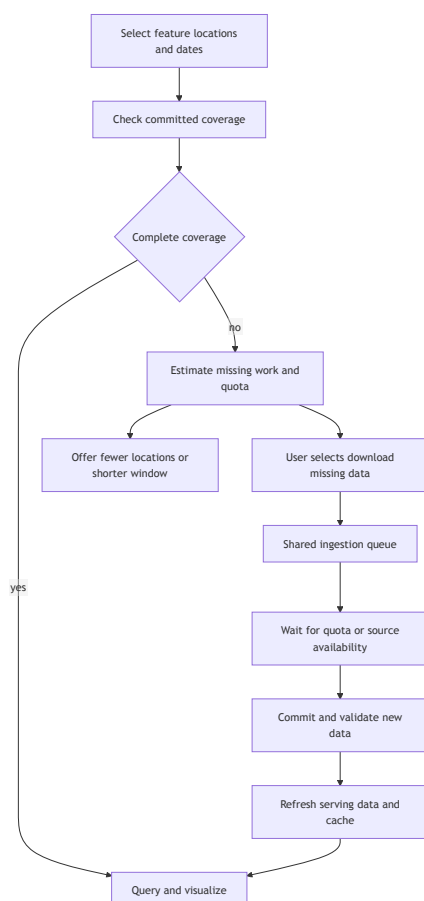


Figure 5.2: Required coverage-aware dashboard workflow.

The Download missing data action must submit only the missing work to the shared ingestion scheduler. It must reuse checkpoints and prevent duplicate requests or competing writers. Requests blocked by the daily budget remain queued with an estimated retry time. Existing data remains available for inspection while they wait. After the required observations are committed and the serving data is refreshed, the dashboard must update coverage, invalidate stale cached results, and rerun the selected analysis.

This workflow belongs to the project scope. The current prototype queries already downloaded data, the coverage suggestions and user-triggered retrieval path remain to be implemented.

5.4 Ask Weather

The Ask weather page accepts a plain-English question and retrieves the current catalogue of features, units, locations, and coverage as model context. The model returns a restricted structured plan. The server validates that plan and compiles an allowlisted query, it does not execute arbitrary model-written Python or SQL. The result includes a chart, plain-text summary, result table, coverage counts, and the exact generated query.

The local demo uses the user's Codex ChatGPT sign-in for planning and summaries, with a separately configured OpenAI API mode available. Spark executes the validated local query. The warehouse design requires an equivalent validated Snowflake query adapter, the existing Spark path must not be described as a deployed warehouse query. Questions are independent and summaries are based on computed aggregates. This is catalogue-grounded analysis rather than a document-vector retrieval index.

Supported requests contain one feature, up to 100 locations, and up to 366 days. They include daily trends, location or country comparisons, elevation scatter plots, and precipitation totals by location. Generated explanations retain units and coverage limitations and must not turn descriptive associations into causal claims.

Ask your weather data

Explore a place, compare elevations, or follow a change over time.

What would you like to see?

Compare rainfall in Colombo and Kandy during August 2026...

Answers use the data downloaded so far.

Explore data

Connected to local Codex · subscription usage applies.

Compare average temperature in Colombo and Kandy over the latest week.

Show daily rainfall in Sri Lanka for the latest available month.

How does temperature vary with elevation across Sri Lanka in the latest week?

Figure 5.3: The Ask Weather dashboard interface

Execution and Reproducibility

6.1 Local Prototype

Install the Python environment and frontend dependencies before starting the explorer. Spark-based analysis requires a compatible Java installation. The raw explorer is served at `http://127.0.0.1:8001`. The main commands are,

```
make install-team
make install-explorer
make regional-plan
make regional-ingest-all
make explorer-raw
```

The default Ask weather demo uses `codex` login with ChatGPT sign-in. Its usage counts against the connected subscription. API mode is selected with `WEATHER_AI_PROVIDER=openai` and a server-side `OPENAI_API_KEY`. Credentials must never be included in browser environment variables or shared report artifacts.

6.2 Warehouse Integration and Release Verification

Warehouse implementation proceeds through role and schema setup, staged loading, dbt source and model configuration, incremental backfill, data-quality tests, and reconciliation with the local reference. Only validated marts should be exposed to the API. On-demand downloads must become queryable through this same validated path.

Release records must retain source coverage, transformation version, configuration, and dataset fingerprint. Evidence should also record the exact query, selected locations, date window, units, and coverage counts. These records make results reproducible across dashboards and separate verified outputs from unfinished work.

6.3 Implementation Constraints and Design Decisions

In practice, the available Snowflake trial allowance of approximately \$400 in free credits was not sufficient for us to complete the MVP. We therefore simulated the Snowflake and dbt process locally using Parquet, PySpark, and versioned release metadata.

The weather API introduced a second constraint, it limited how much data could be extracted in a single request. Requesting the entire historical range for every location and variable was therefore impractical and increased the risk of failed or incomplete downloads. The ingestion strategy was changed to prioritise the most recent available data across the study locations. Historical data can then be requested and streamed into the application on demand when a user selects a date range that is not already available locally. This approach gives users a useful current view quickly, while avoiding an unnecessarily large initial extraction.